

Vector solitons in nonlinear lattices with competing SPM and XPM terms

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We demonstrate that two-component photonic nonlinear lattices with opposite signs of the out-of-phase spatially modulated SPM and XPM coefficients can support two types of optical vector solitons (VSs), the fundamental and mixed ones. The fundamental VSs are composed of two Gaussian-like components, while the mixed ones combine Gaussian-like and dipole components. The structures and stability of both types of the VSs are investigated. The VSs are stable, except for narrow intervals at small values of the propagation constant.

Keywords: Vector solitons; Self-phase modulation; Cross-phase modulation; Nonlinear Optics

I. INTRODUCTION

The generation of solitons is a basic topic in many areas of nonlinear physics [1–14]. In particular, studies of solitons have drawn much interest in nonlinear optics [15–23] and Bose-Einstein condensates (BECs) [24–31]. Generally, the formation and evolution of solitons is governed by diverse forms of the nonlinear Schrödinger equation (NLSE). Various types of solitons have been produced with the help of spatially periodic linear [25] and nonlinear [10] potentials, alias linear and nonlinear lattices. These are gap solitons, populating finite bandgaps in the spectrum induced by the linear lattice [32–36], and solitons created by nonlinear lattices, including fundamental [37–40], dipole [41], multipole [42, 43], dark [44, 45] and vortex [46] states.

Vector solitons (VSs) are modes which contain two or more components [8, 12, 47–49], modeled by systems of coupled NLSEs. In particular, VSs may carry different propagation constants (chemical potentials, in terms of BEC) in their two components. Many types of VSs have been investigated [50–54], such as fundamental ones [55], dipoles [56, 57], “soliton molecules” (bound states) [58], ring solitons [59], and vortices [60]. In particular, VSs of the gap type in systems combining the saturable nonlinearity and linear-lattice potentials have been reported [61, 62]. Although the vector solitons in periodic linear potentials have been widely investigated, the ones under the action of periodically patterned self-phase modulation (SPM) and cross-phase modulation (XPM), in the absence of a linear lattice, call for more attention, especially to VSs built of components with different forms. Such nonlinear lattices and states supported by them may be realized in photorefractive crystals.

In this work, we study two-component VSs in nonlinear lattices (without any linear potential). Two types of VSs are produced, namely, the fundamental and mixed ones. The former species is built as a single-peak mode with Gaussian-like shapes of its components, while the mixed solitons combine fundamental and dipole components.

The presentation is arranged as follows. We report the model and linear-stability analysis of the VSs in Section II, where we also provide a qualitative explanation of the possibility of the formation of stable VSs in the system. Numerical results are reported in Section III, which is divided in two parts, that address the fundamental and mixed VSs, respectively. The paper is concluded by Section IV.

II. THE MODEL

The propagations of optical VSs in the nonlinear lattice are modeled by the system of coupled NLSEs, written in terms of the scaled propagation distance z and transverse coordinate x :

$$\begin{aligned} i \frac{\partial E_1}{\partial z} &= -\frac{1}{2} \frac{\partial^2 E_1}{\partial x^2} + [G(x)|E_1|^2 + \Gamma(x)|E_2|^2] E_1, \\ i \frac{\partial E_2}{\partial z} &= -\frac{1}{2} \frac{\partial^2 E_2}{\partial x^2} + [G(x)|E_2|^2 + \Gamma(x)|E_1|^2] E_2. \end{aligned} \quad (1)$$

Variables $E_{1,2}$ are the slowly varying amplitudes of the two components, which may represent orthogonal polarizations of light or optical waves with different carrier wavelengths. The nonlinear lattice is introduced by adopting spatially-modulated

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expressions,

$$G(x) = -\cos^2 x, \quad \Gamma(x) = \gamma \sin^2 x, \quad (2)$$

for the local SPM and XPM coefficients, with $\gamma > 0$, which makes the SPM and XPM signs opposite, corresponding to the competition between the focusing and defocusing nonlinearities. The lattice period in Eq. (2) is fixed to be π by means of scaling. We consider the systems with $\gamma = 2/3$ and 2, which correspond, respectively, to the XPM interaction between mutually orthogonal linear polarizations of light, or the interaction between left- and right-twisted circular polarizations of light, as well as the between optical waves carried by different wavelengths [17]. In fact, values of γ between $2/3$ and 2 produce results which are quite similar to those reported below. The opposite SPM and XPM signs are possible in photonic crystals [63, 64], in which the spatial modulation can be fabricated by means of available technologies [65, 66]. The system of equations (1) conserves three dynamical invariants: powers of the two components,

$$P_{1,2} = \int_{-\infty}^{+\infty} |E_{1,2}(x)|^2 dx, \quad (3)$$

and the Hamiltonian,

$$H = \int_{-\infty}^{+\infty} \left\{ \frac{1}{2} \sum_{n=1,2} \left[\left| \frac{\partial E_n}{\partial x} \right|^2 + G |E_n|^4 \right] + \Gamma |E_1|^2 |E_2|^2 \right\} dx. \quad (4)$$

Stationary solutions for the VSs are produced by substituting

$$E_{1,2} = U_{1,2}(x) \exp(ib_{1,2}z) \quad (5)$$

in Eq. (1), where $b_{1,2} > 0$ are propagation constants of the two components. This leads to the system of stationary equations for real wave functions $U_{1,2}$,

$$\begin{aligned} -b_1 U_1 &= -\frac{1}{2} \frac{d^2 U_1}{dx^2} + [G(x)U_1^2 + \Gamma(x)U_2^2] U_1, \\ -b_2 U_2 &= -\frac{1}{2} \frac{d^2 U_2}{dx^2} + [G(x)U_2^2 + \Gamma(x)U_1^2] U_2. \end{aligned} \quad (6)$$

In the spatially uniform system, with constant coefficients $G < 0$ and $\Gamma > 0$, VSs in the form of a bound state of two fundamental components E_1 and E_2 cannot be stable, as $\Gamma > 0$ implies repulsion between the components, hence the two-component soliton would be unstable against spontaneous splitting into separating single-component ones. Nevertheless, the nonlinear lattice structure, adopted in Eq. (2), makes the existence of a stable two-component mode possible. To demonstrate this possibility in an approximate form, one can treat the XPM terms in Eqs. (1) and (6) as small perturbations, which is possible if γ in Eq. (2) is a small coefficient, or the power of one soliton is much smaller than the other's, $P_2 \ll P_1$ (in other words, $b_2 \ll b_1$ in Eqs. (6)). Then, in the zeroth-order approximation, the E_1 component is shaped as a fundamental soliton, $(U_1)_{\text{sol}} = 2\sqrt{b_1} \text{sech}(\sqrt{2b_1}x)$, if the spatially-modulated SPM coefficient $G(x)$ is replaced by its average value, $\bar{G} = -1/2$ (see Eq. (2)). Finally, the XPM term in the second equation of system (1) gives rise to an effective potential for field E_2 :

$$W_{\text{eff}}(x) = 4b_1\gamma \frac{\sin^2 x}{\cosh^2(\sqrt{2b_1}x)}, \quad (7)$$

where $\Gamma(x)$ is substituted as per Eq. (2). Expression (7) describes a *potential quasi-well* centered at $x = 0$. In the linear limit, the quasi-well supports leaky (gradually decaying) bound states, as the potential, vanishing at $|x| \rightarrow \infty$, is positive at all x . Nevertheless, the inclusion of the self-focusing nonlinearity readily transforms the leaky bound states into truly bound ones [67]. In other words, the potential quasi-well may provide stable trapping of a soliton in the E_2 component, thus building a genuine robust VS.

As concerns the bound state of the fundamental mode in E_1 and the dipole one in E_2 , the latter component may be considered, in a crude approximation, as a pair of out-of-phase solitons trapped in the nonlinear lattice induced by coefficient $G(x)$ (see Eq. (2)), which interact with the fundamental mode in component E_1 through the effective potential (7).

To explore the stability of the VSs against small perturbations, we disturb the VS solution as

$$E_{1,2}(x, z) = [U_{1,2}(x) + p_{1,2}(x)e^{\lambda z} + q_{1,2}^*(x)e^{\lambda^* z}]e^{ib_{1,2}z}, \quad (8)$$

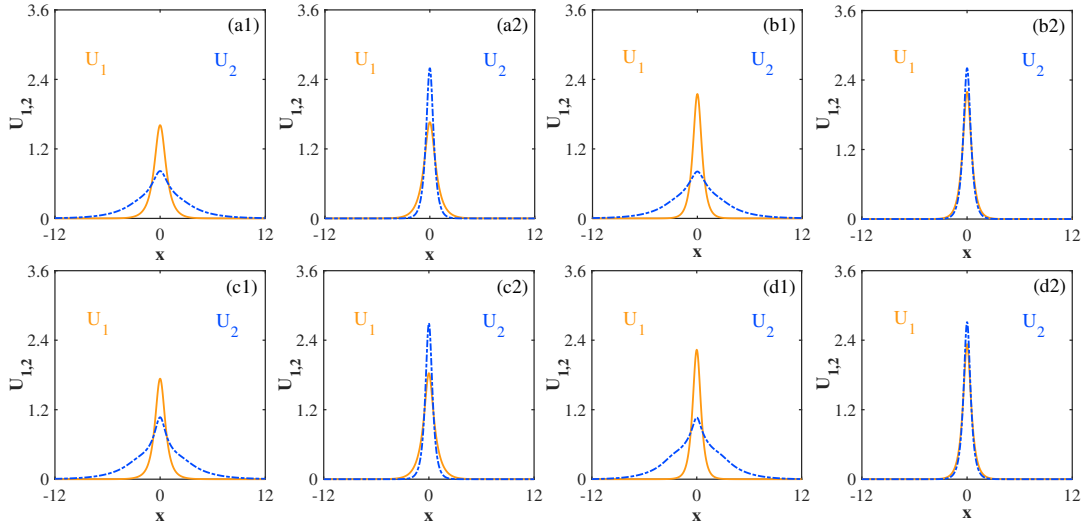


FIG. 1. Profiles of fundamental VSs in the first and second components (yellow and blue lines, respectively) for $\gamma = 2/3$: (a1) unstable one at $b_1 = 1, b_2 = 0.1$; (a2) stable one at $b_1 = 1, b_2 = 3$; (b1) unstable one at $b_1 = 2, b_2 = 0.1$; (b2) stable one at $b_1 = 2, b_2 = 3$. Panels (c1)–(d2): the same as in (a1)–(b2) but for $\gamma = 2$.

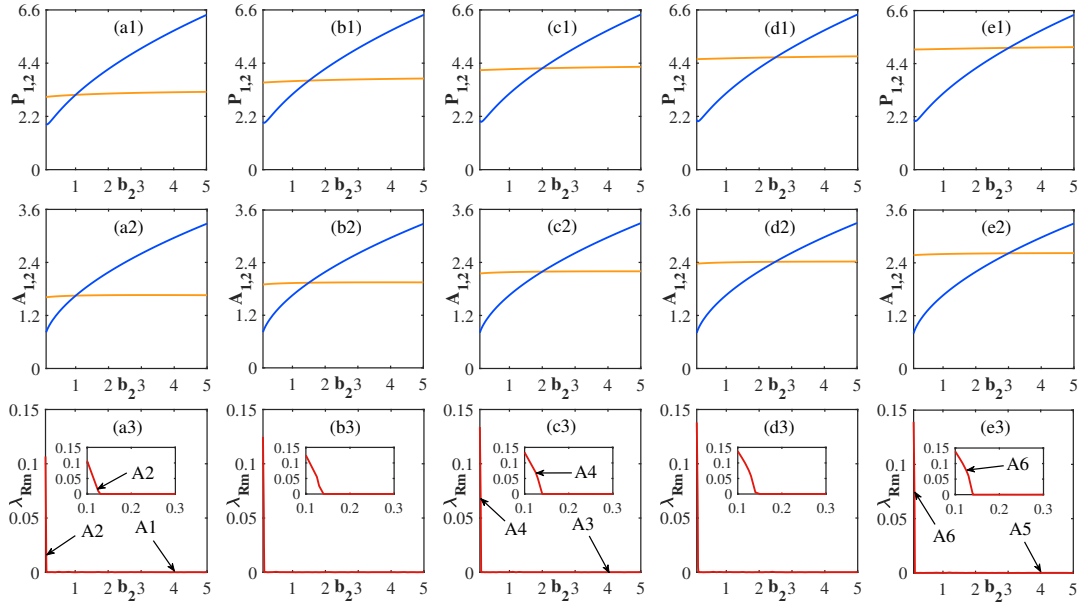


FIG. 2. Powers $P_{1,2}$ (top row), amplitudes $A_{1,2}$ (middle row), and maximum real parts of eigenvalues λ_{Rm} (bottom row) of fundamental VSs versus the propagation constant of the second component, b_2 , for $\gamma = 2/3$: (a1)–(a3) with $b_1 = 1$; (b1)–(b3) with $b_1 = 1.5$; (c1)–(c3) with $b_1 = 2$; (d1)–(d3) with $b_1 = 2.5$; (e1)–(e3) with $b_1 = 3$. The yellow and blue lines pertain to the first and second components, respectively.

where λ (which may be complex) is the instability growth rate corresponding to the perturbation eigenmode with components $p_{1,2}(x)$ and $q_{1,2}^*(x)$, the asterisk standing for the complex conjugate. The substitution of ansatz (8) into Eq. (1) and linearization with respect to the perturbation leads to the following eigenvalue problem for λ :

$$i\lambda \begin{pmatrix} p_1 \\ q_1 \\ p_2 \\ q_2 \end{pmatrix} = \begin{pmatrix} \hat{L}_1 & \hat{W}_1 & \hat{W}_2 & \hat{W}_2 \\ -\hat{W}_1 & -\hat{L}_1 & -\hat{W}_2 & -\hat{W}_2 \\ \hat{W}_2 & \hat{W}_2 & \hat{L}_2 & \hat{W}_3 \\ -\hat{W}_2 & -\hat{W}_2 & -\hat{W}_3 & -\hat{L}_2 \end{pmatrix} \begin{pmatrix} p_1 \\ q_1 \\ p_2 \\ q_2 \end{pmatrix}, \quad (9)$$

where

$$\begin{aligned}\hat{L}_1 &= -(1/2)(\partial^2/\partial x^2) + b_1 + 2GU_1^2 + \Gamma U_2^2, \\ \hat{L}_2 &= -(1/2)(\partial^2/\partial x^2) + b_2 + 2GU_2^2 + \Gamma U_1^2, \\ \hat{W}_1 &= GU_1^2, \hat{W}_2 = \Gamma U_1 U_2, \hat{W}_3 = GU_2^2.\end{aligned}\quad (10)$$

Obviously, the stationary solutions are stable if $\text{Re}(\lambda) = 0$ for all eigenvalues.

Below, the stationary VS solutions of Eq. (6) are obtained by means of the modified squared-operator method [68]. The stability of VSs is identified by the spectrum of eigenvalues λ and is corroborated by numerical simulations of the perturbed evolution, which are performed using the standard finite-difference method. In this work, the computational domain for the fundamental VSs is $x \in [-15, 15]$, while the one for the mixed VSs is $x \in [-20, 20]$. The spatial and propagation steps are $dx = 0.05$ and $dz = 0.00025$, respectively. The convergence tolerance is 10^{-10} . We use the Laplacian-zero (neglecting the term of Laplacian second derivative) [69] as the boundary condition in this work. The perturbation amplitude is 1% of the stationary solution. And we employ the Fourier collocation method [70] as the eigensolver.

III. NUMERICAL RESULTS

In this section, we focus on the numerical results for the VSs, including profiles, power, amplitude, stability and perturbed propagations of the solitons. Two types of VSs are investigated in this section, namely, the fundamental and mixed VSs. In the former case, both components are fundamental modes. In the mixed VS, one component remains fundamental, while the other one features a dipole shape. The results for the fundamental and mixed VSs are presented here in subsections III A and III B, respectively.

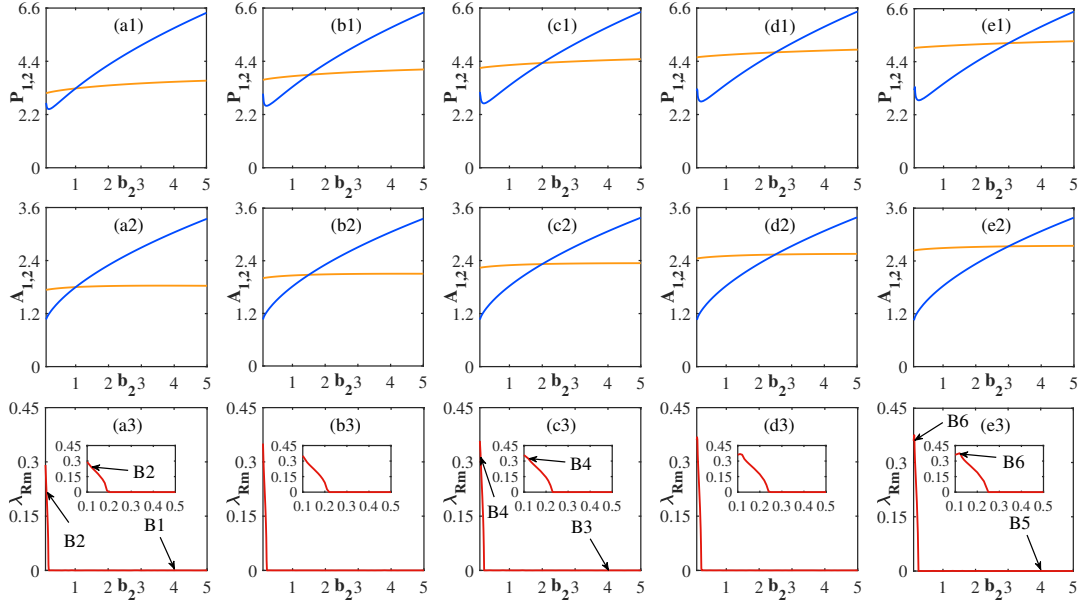


FIG. 3. The same as in Fig. 2, but with $\gamma = 2$.

A. Fundamental VSs

For the physically relevant value of the XPM coefficient $\gamma = 2/3$, typical profiles of unstable fundamental VSs with widely different propagation constants of the first and second components, *viz.*, $b_1 = 1, b_2 = 0.1$ and $b_1 = 2, b_2 = 0.1$ are presented in Figs. 1(a1) and (b1), respectively. In this case, the second equation in system (6) suggests that the width of the second component scales as $b_2^{-1/2}$, *i.e.*, it is relatively large when b_2 is small and, naturally, the amplitude of this component is smaller than its counterpart in the first component, $A_2 < A_1$. On the other hand, when the values of the propagation constants take values on the same order of magnitude, *viz.*, $b_1 = 1, b_2 = 3$ and $b_1 = 2, b_2 = 3$ in Figs. 1(a2) and (b2), respectively, the widths

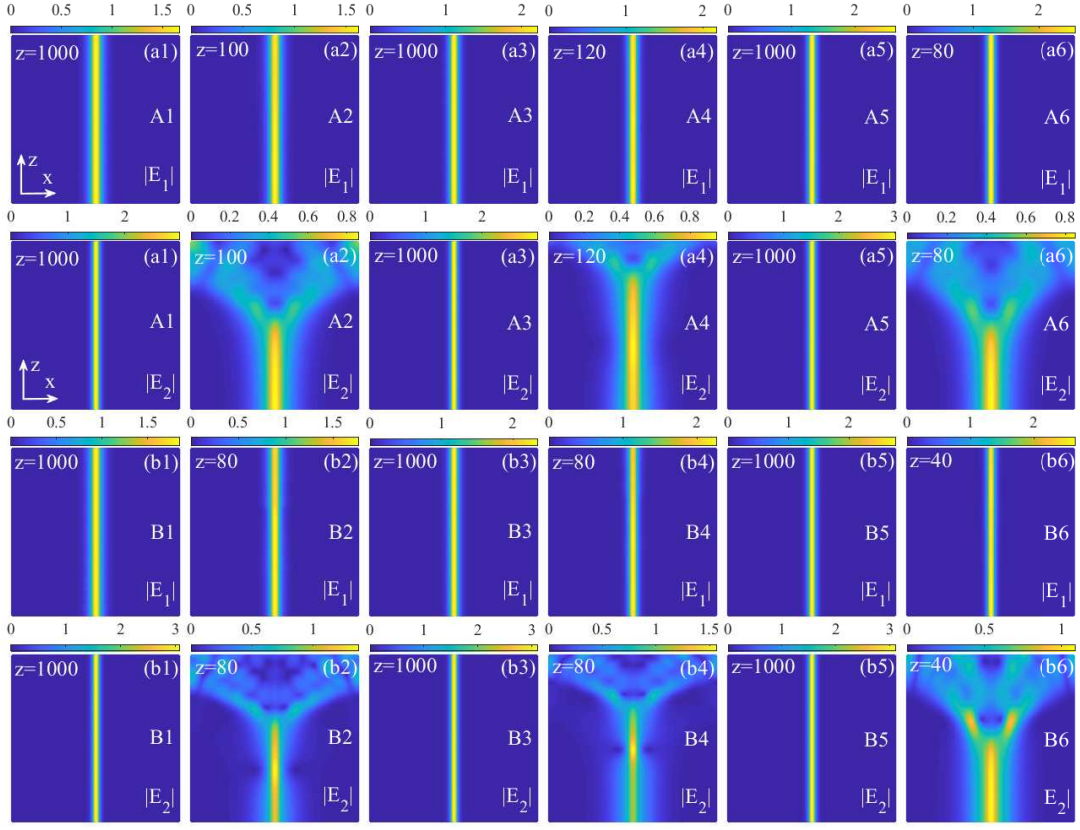


FIG. 4. Perturbed propagations of fundamental VSs at $\gamma = 2/3$: (a1) with $b_1 = 1$, $b_2 = 4$; (a2) with $b_1 = 1$, $b_2 = 0.125$; (a3) with $b_1 = 2$, $b_2 = 4$; (a4) with $b_1 = 2$, $b_2 = 0.125$; (a5) with $b_1 = 3$, $b_2 = 4$; (a6) with $b_1 = 3$, $b_2 = 0.125$. Panels (b1) - (b6) show the same for $\gamma = 2$. In all panels, the simulation domain is $|x| \leq 15$.

of the two components are nearly equal, and the amplitude A is larger for larger b . Figures 1(c1)–(d2) present similar results, but for the other physically relevant value of the XPM coefficient, $\gamma = 2$ (recall that it corresponds to the circular polarizations of light or different wavelengths).

Next, we systematically address the power, amplitude and stability of the fundamental-VS family. These characteristics for the XPM coefficient γ are presented, severally, in the top, middle and bottom rows of Figs. 2 and 3, for $\gamma = 2/3$ and 2, respectively, as functions of b_2 , at several fixed values of the propagation constant of the first component, viz., $b_1 = 1, 1.5, 2, 2.5$. It is seen that power P_2 always increases with the increase of b_2 , while power P_1 increases very slowly, almost keeping a constant value. Further, the amplitude of the second component, A_2 , increases with the increase of b_2 , while A_1 remains almost constant.

The maximum real part λ_{Rm} of the instability growth rate [given by Eqs. (9)] versus b_2 with the values of $b_1 = 1, 1.5, 2, 2.5, 3$ of fundamental VSs at $\gamma = 2/3$ are portrayed in Figs. 2(a3)–(e3), respectively. According to Figs. 2(a3)–(e3), these fundamental VSs become unstable ($\lambda_{Rm} > 0$) when b_2 is very small (in the left parts of the panels). And the curves of λ_{Rm} versus b_2 for such unstable VSs can be clearly seen in the subgraphs of these panels. The perturbed propagations of the fundamental VSs labelled by A1–A6 are reported in Figs. 4(a1)–(a6), respectively.

The power, amplitude and linear stability analysis of the fundamental VSs at $\gamma = 2$ are presented in the top, middle and bottom rows of Fig. 3, respectively. Specifically, the power of both components P_1 and P_2 versus b_2 with values of $b_1 = 1, 1.5, 2, 2.5, 3$ are reported in Figs. 3(a1)–(e1) from which one can see that P_1 (yellow line) and P_2 (blue line) always increase with the increase of b_2 . And the increasing speed of P_1 versus b_2 at $\gamma = 2$ is slower than the one of P_2 . In addition, P_1 increase with the increase of b_1 obviously. Next we present the amplitude of both components versus b_2 in Figs. 3(a2)–(e2) where both A_1 and A_2 increase with the increase of b_2 and the increasing speed of A_2 versus b_2 is faster than the one of A_1 . Further, we can also find that A_1 increase with b_1 obviously.

The dependence of the maximum real part λ_{Rm} of the instability growth rate [produced by the numerical solution of Eqs. (9)] versus b_2 with the values on b_2 demonstrates that the fundamental VSs are unstable ($\lambda_{Rm} > 0$) only at very small values of b_2 . To verify the linear-stability predictions, simulations of the perturbed propagation of the fundamental VSs are presented in Fig. 4. As expected, the VSs with $\lambda_{Rm} = 0$, labelled A1, A3, A5, B1, B3, B5 in Fig. 4 remain stable up to the largest propagation

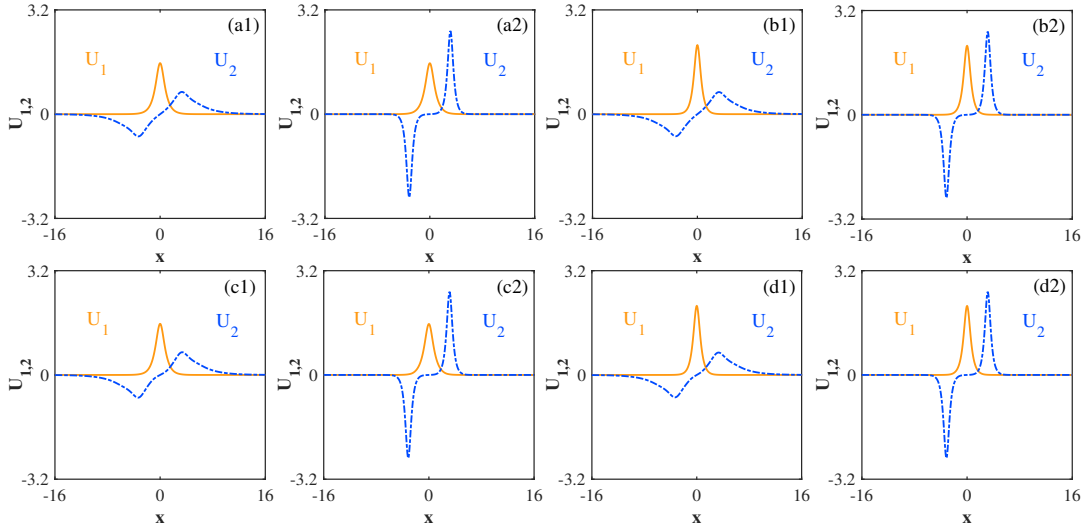


FIG. 5. Profiles of mixed VSs in the first and second components (yellow and blue lines, respectively) with the XPM parameter $\gamma = 2/3$: (a1) unstable one at $b_1 = 1, b_2 = 0.1$; (a2) stable one at $b_1 = 1, b_2 = 3$; (b1) unstable one at $b_1 = 2, b_2 = 0.1$; (b2) stable one at $b_1 = 2, b_2 = 3$. Panels (c1)–(d2): the same as (a1)–(b2), but for $\gamma = 2$.

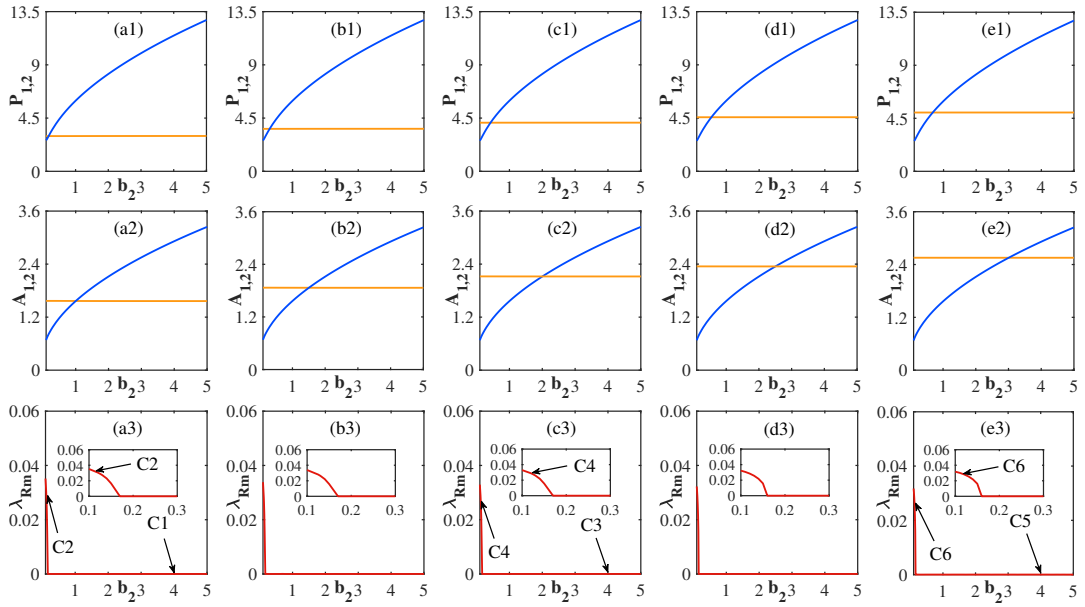


FIG. 6. Powers $P_{1,2}$ (the top row), amplitudes $A_{1,2}$ (the middle row), and the maximum real part of eigenvalues λ_{Rm} (the bottom row) of mixed VSs, for $\gamma = 2/3$, versus the propagation constant b_2 of the dipole component b_2 : (a1)–(a3) for $b_1 = 1$; (b1)–(b3) for $b_1 = 1.5$; (c1)–(c3) for $b_1 = 2$; (d1)–(d3) for $b_1 = 2.5$; (e1)–(e3) for $b_1 = 3$. Yellow and blue lines display the results of the fundamental and dipole components, respectively.

distance tested here, $z = 1000$. On the other hand, the fundamental VSs labelled A2, A4, A6, B2, B4, B6, which correspond to $\lambda_{Rm} > 0$, are indeed unstable, exhibiting sudden decay of the amplitude of the second component, thus spontaneously transforming into obviously stable single-component solitons.

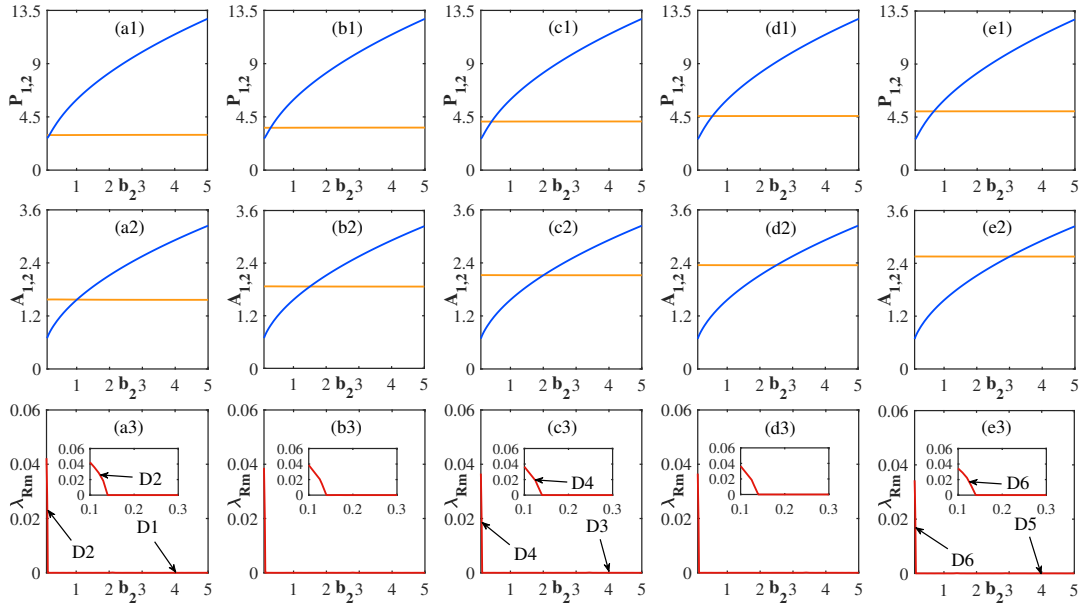


FIG. 7. The same as in Fig. 6, but for $\gamma = 2$.

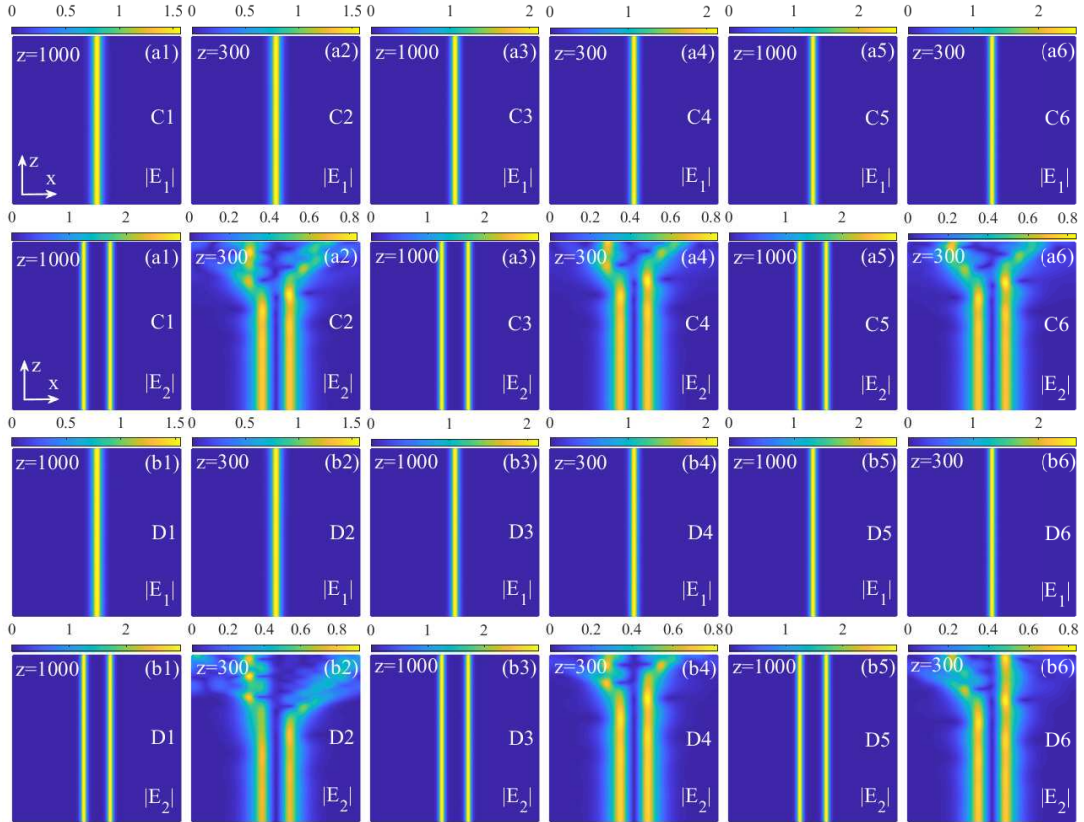


FIG. 8. The perturbed propagations of mixed VSs for $\gamma = 2/3$: (a1) with $b_1 = 1, b_2 = 4$; (a2) with $b_1 = 1, b_2 = 0.125$; (a3) with $b_1 = 2, b_2 = 4$; (a4) with $b_1 = 2, b_2 = 0.125$; (a5) with $b_1 = 3, b_2 = 4$; (a6) with $b_1 = 3, b_2 = 0.125$. Panels (b1) - (b6) show the same for $\gamma = 2$. In all panels, the simulation domain is $|x| \leq 20$.

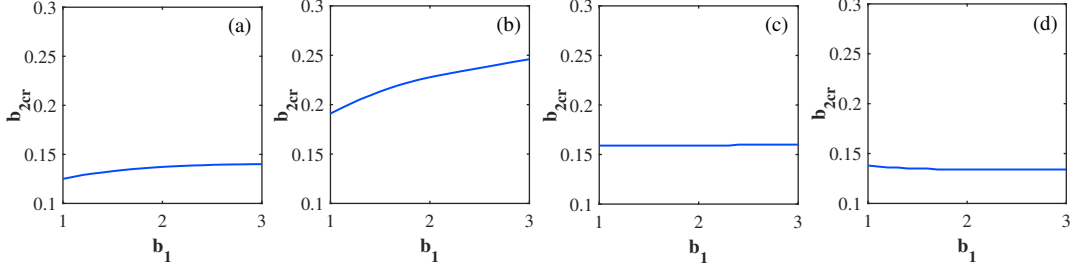


FIG. 9. The critical value of b_2 versus b_1 for these soliton families: (a) fundamental VSs at $\gamma = 2/3$; (b) fundamental VSs at $\gamma = 2$; (c) mixed VSs at $\gamma = 2/3$; (d) mixed VSs at $\gamma = 2$.

B. Mixed VSs

Recall that the mixed VSs are composed of two different components, fundamental and dipole ones. Profiles of the mixed VSs with small and relatively large values of the dipole's propagation constants b_2 are presented in Figs. 5(a1,b1,c1,d1) and (a2,b2,c2,d2), respectively. In the former case, the dipole's amplitude is small, and its shape is smooth, while in the latter case the amplitude is much higher, and the dipole effectively splits in a pair of separated peaks with opposite signs.

Dependences of the power, amplitude and linear stability of the mixed-VS family on the propagation constant b_2 of the dipole component, for $\gamma = 2/3$ and $\gamma = 2$, are plotted in the top, middle, and bottom rows of Figs. 6 and 7, respectively. In both figures, the dependences are produced for fixed values of the propagation constant of the fundamental component, $b_1 = 1.0, 1.5, 2.0, 2.5, 3.0$, in columns (a)–(e), respectively.

In the same Figs. 6 and 7, panels (a3)–(e3) display dependences of the maximum real part λ_{Rm} of the instability growth rate versus b_2 . It is seen that, as well as the fundamental VSs [see panels (a3)–(e3) in Figs. 2 and 3], the mixed ones have instability intervals only at very small values of b_2 (in the left parts of the panels). We also present the curves of λ_{Rm} versus b_2 with the small value of b_2 (unstable mixed VSs) in the subgraphs of the panels (a3)–(e3).

The predictions for the stability of mixed VS, produced by the linear-stability analysis, is verified by direct simulations of the perturbed propagation in Fig. 8. It is seen that the mixed VSs labelled by C1, C3, C5, D1, D3, D5 in Fig. 8 are stable, while ones labelled by C2, C4, C6, D2, D4, D6 are unstable. Similar to their counterparts of the fundamental type (see Fig. 4), the dipolar components of the unstable mixed VSs get spontaneously destroyed, while the fundamental components survive as single-component solitons.

Finally, to investigate the stability domains of the soliton families, relations between the critical value of b_2 (b_{2cr}) and b_1 , for both fundamental and mixed VSs, are presented in Fig. 9. Specifically, the curves of $b_{2cr}(b_1)$ for the fundamental VSs, with $\gamma = 2/3$ and $\gamma = 2$, are plotted in Figs. 9(a) and 9(b), while the ones for mixed VSs, with $\gamma = 2/3$ and $\gamma = 2$, are plotted in Figs. 9(c) and 9(d). Obviously, in panel (b) the value of b_{2cr} is sensitive to the value of b_1 , while the variation of b_{2cr} with the increase of b_1 is quite weak in other panels.

IV. CONCLUSION

In this work, we have demonstrated that nonlinear photonic lattices (in the absence of the linear potential) with opposite signs of the SPM and XPM terms sustain two types of optical VSs (vector solitons), *viz.*, the fundamental and mixed ones. The fundamental VSs are built of two fundamental components, while the mixed ones combined fundamental and dipole components. The shapes, powers, amplitudes and linear stability of both VS types are systematically investigated. In particular, the dependences of these characteristics of the mixed VS family on the propagation constant of the dipolar component are reported. The stability domains of these VSs, predicted by the linear-stability analysis, have been verified by direct simulations of the perturbed propagation. The VSs are unstable only at very small values of the propagation constant of one component (the dipolar one, in the case of mixed VSs).

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DATA AVAILABILITY STATEMENT

Data will be made available on a reasonable request.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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